NetDoctor

NETWORK DIAGNOSTIC SYSTEM — PHASE 2 COMPLETION REPORT

Phase Completed: Phase 2 (Problem Detection & Validation)	**Status:** 100% Implemented & Verified
Test Suites: 32/32 Tests Passed across 3 test suites	**Target Categories:** 11 MVP Categories Recognized

1. Phase 2 Implementation Summary

Phase 2 delivers a deterministic, explainable natural-language problem understanding and validation pipeline for NetDoctor. Before any diagnostic tests are performed, the system validates the problem scope, rejects non-networking issues, tolerates spelling errors, asks single targeted follow-up questions for vague queries, and outputs a structured Validated Problem Context ready for Phase 3.

2. Supported 11 MVP Problem Categories

Category Key	Display Name	Sample Natural Language Input
`internet_not_working`	Internet Not Working	"My internet is completely down and offline"
`wifi_connection`	Wi-Fi Connection Problems	"WiFi keeps disconnecting from router"
`slow_internet`	Slow Internet Throughput	"The internet speed is extremely slow and buffering"
`high_latency`	High Ping & Latency	"My ping is very high and gaming lag is bad"
`packet_loss`	Packet Loss	"Getting 20% packet loss in Discord calls"
`dns_problem`	DNS Resolution Failure	"DNS server not responding, cannot resolve domain"
`ip_dhcp_problem`	IP & DHCP Issues	"Computer is stuck with 169.254 self-assigned IP"
`router_connectivity`	Router & Gateway	"Cannot reach my default gateway at 192.168.1.1"
`network_adapter`	Network Adapter Issues	"My Wi-Fi network adapter isn't working or recognized"
`website_connectivity`	Specific Website Issue	"One specific website won't open but others load"
`lan_connectivity`	LAN / Local Network	"Cannot connect to another computer on my LAN"

3. Validation, Vague Queries & Typo Tolerance

- **Non-Networking Rejection:** Queries such as *'My Java code has an error'*, *'Laptop battery is draining'*, *'How do I cook pasta?'*, and *'Printer out of ink'* are rejected with a beginner-friendly alert without entering the workspace.
- **Vague Query Handling:** Generic inputs (*'Internet not working'*, *'Network problem'*, *'Net is gone'*, *'WiFi problem'*) trigger a single targeted clarification question (*'Is the problem occurring on one device or multiple devices?'*).
- **Typo & Fuzzy Tolerance:** Misspellings like disconecting → disconnecting, internat → internet, pign → ping, wierless → wireless, and workng → working are corrected automatically.
- **Compound Disambiguation:** Complex queries like *'WiFi is connected but websites will not open'* correctly prioritize DNS/Internet resolution rather than physical Wi-Fi link drops.

4. Automated Test Verification Summary

Test Suite File	Coverage Scope	Status
`Phase2Classification.test.ts`	11 Categories, Non-Networking, Typos, Vague queries, Context Handoff	**23/23 PASSED**
`Phase2UI.test.tsx`	Rejection Alerts, Follow-Up Card UI, Typo Workspace Transition	**3/3 PASSED**
`Phase1UI.test.tsx`	Regression Test Suite: Branding, Roadmap, Step Navigation, Modals	**6/6 PASSED**
Production Build (tsc -b)	Full client build validation	**0 ERRORS**

5. Problem Context Package Structure (Phase 3 Handoff)

Phase 2 outputs a strongly-typed ValidatedProblemContext object containing: originalDescription, normalizedDescription, category (key), categoryDisplayName, operatingSystem, followUpAnswer, isOSSupported, and contextSummary. Phase 3 (Structured Diagnosis & Troubleshooting) will consume this context directly to generate custom roadmap procedures.